

CS 111

Foundations of Program Design
Week 3 Lecture

Quiz 2

This lecture

1. Conditional blocks
2. FizzBuzz
3. Quiz
4. – break –
5. `git fetch / pull`
6. While loops
7. Slug generator

Additional course resources

The University of Helsinki offers a free online Java textbook. We are about to have covered Part 1:

<https://java-programming.mooc.fi/part-1>

As we go forward, slides will reference appropriate `mooc.fi` sections that you may review.

Block Statements

A code block is enclosed by a pair of curly brackets `{}`.

```
// the class definition is a block
public class Echo {

}
```

Block Statements

A code block is enclosed by a pair of curly brackets `{}`.

```
// the class definition is a block
public class Echo {

    // ...as is the main() entry point method
    public static void main(String[] args) {

    }
}
```

Block Statements

A code block is enclosed by a pair of curly brackets `{}`.

```
// the class definition is a block
public class Echo {

    // ...as is the main() entry point method
    public static void main(String[] args) {

        // ...as is the conditional statement!
        if (args.length ≥ 1) {
            System.out.println(args[0]);
        }
    }
}
```

Note the `args[0]` array indexing syntax, which is new to us.

So is the `args.length` property provided by the string array, `String[] args`.

Let's explore further during lab.

Quiz 2

Followed by a break, then lab

Compile and run `java Echo hello world`. Only `hello` will print.

```
// the class definition is a block
public class Echo {

    // ...as is the main() entry point method
    public static void main(String[] args) {

        // ...as is the conditional statement!
        if (args.length ≥ 1) {
            System.out.println(args[0]);
        }
        else {
            System.out.println("I need to see an argument");
        }
    }
}
```

1. What is in `args[1]` ?
2. What is the `args.length` guard doing?

Create a new private repository in the `CS111-jtotto` org, titled `YourName-Workbench` . Then clone it to your machine:

<https://github.com/orgs/CS111-jtotto/repositories>

Make an empty file `FizzBuzz.java` in this repository, and push it to GitHub.

If the SSH pull fails, follow steps in `Week 1/ssh_agent.md` matching your OS, using your GitHub email to generate your SSH key pair.

Bonus: Compile and run `java SlugGenerator`. Note the new `import` syntax.

Human-readable slugs are used to uniquely name instances of e.g. a shared virtual room.

```
import java.util.Random;

public class SlugGenerator {
    public static void main(String[] args) {
        String[] adjectives = {"Brave", "Clever", "Daring", "Eager",
                               "Fierce", "Gentle", "Happy", "Jumpy", "Quick"};
        String[] animals    = {"Badger", "Cat", "Dolphin", "Eagle",
                               "Fox", "Gecko", "Hippo", "Jaguar", "Quail"};

        Random random = new Random();

        // Since the adjectives and animals are parallel arrays,
        // they will always start with the same letter!
        int index = random.nextInt(adjectives.length);

        String slug = adjectives[index] + "-" + animals[index];
        System.out.println(slug);
    }
}
```

1. Mix up the adjectives and animals; right now, they always start with the same character.
2. Count the number of unique slugs possible with the current generator versus your modified version.